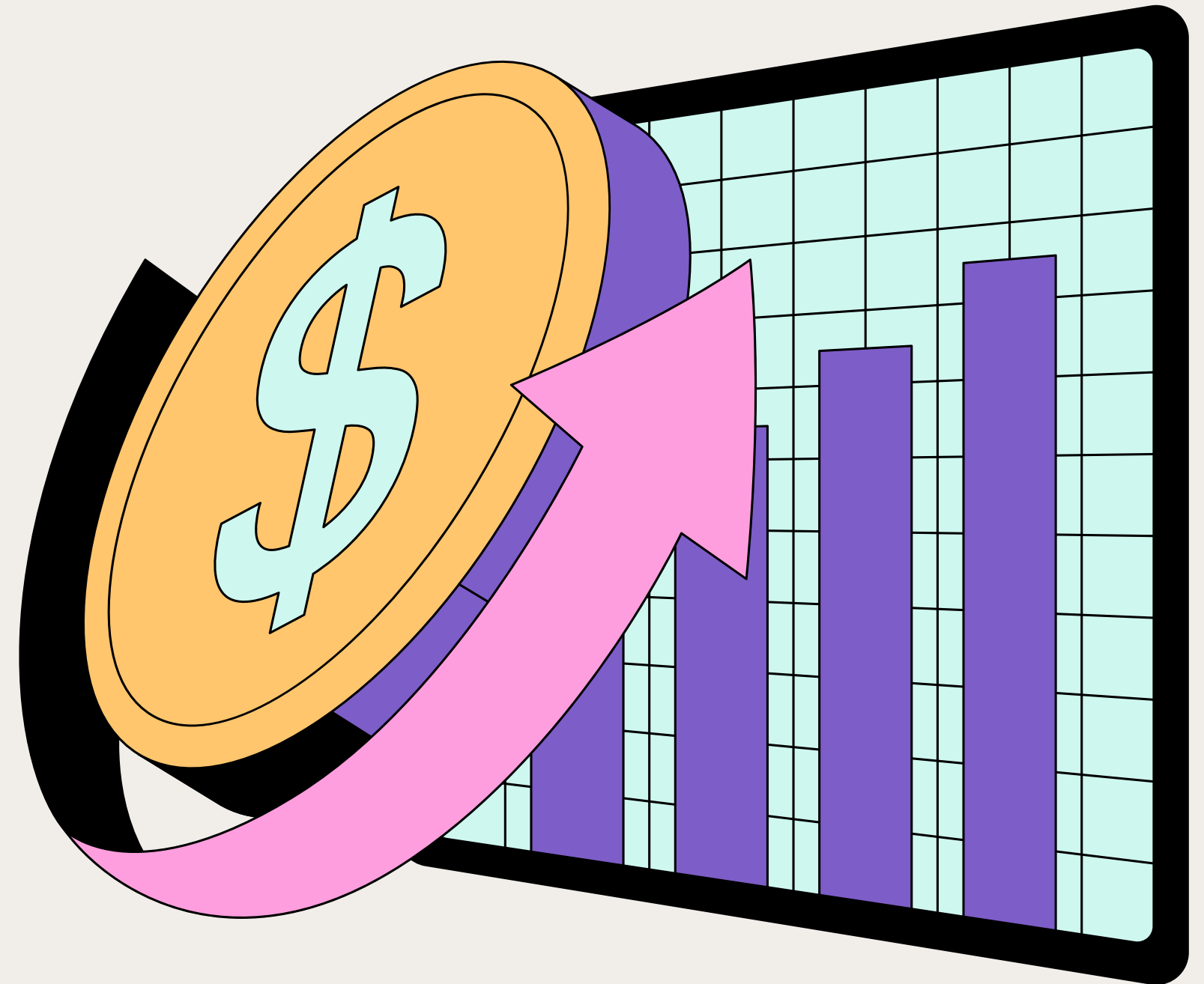


Super Store

Optimalisasi Profit dengan Machine Learning

Marcella Aurelia Yatijan - 71477



Tujuan

Memprediksi Profit dan Rugi
Order Berdasarkan Pola
Historis untuk Optimasi
Keputusan Bisnis

Mengetahui variabel yang
paling berpengaruh
terhadap profit



Profit dan Discount Bin

Kelas	Syarat	Penjelasan
'Rugi'	$p < 0$	Jika profit negatif (rugi)
'Untung Kecil'	$0 \leq p \leq 30$	Jika profit sedikit
'Untung Besar'	$p > 30$	Jika profit besar

=== Statistik Profit ===

Mean : 28.64

Median : 9.24

Modus : [0.0]

discount_bin	Range Diskon (discount)	Penjelasan
0	0.00	Tanpa diskon
1	>0.00 s.d. 0.10	Diskon rendah (0–10%)
2	>0.10 s.d. 0.30	Diskon sedang (10–30%)
3	>0.30 s.d. max_val (mungkin 0.8–1.0)	Diskon tinggi (>30%)



Feature Importance by Chi Squared

===== Hasil Uji Chi-Squared terhadap 'profit_bin' =====					
Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan	
discount_bin	36065.6884	12.5916	0.0000	✓	Tolak H0
product_name	28422.4752	7777.5746	0.0000	✓	Tolak H0
country	24149.4983	332.8538	0.0000	✓	Tolak H0
sub_category	10604.9665	46.1943	0.0000	✓	Tolak H0
category	6220.4128	9.4877	0.0000	✓	Tolak H0
ship_date	4812.2120	3052.9553	0.0000	✓	Tolak H0
region	3323.9449	36.4150	0.0000	✓	Tolak H0
customer_name	2643.4157	1681.8204	0.0000	✓	Tolak H0
market	1397.6546	21.0261	0.0000	✓	Tolak H0
order_priority	4.2271	12.5916	0.6460	Gagal	Tolak H0
ship_mode	3.1470	12.5916	0.7902	Gagal	Tolak H0
segment	1.0461	9.4877	0.9027	Gagal	Tolak H0



FEATURE

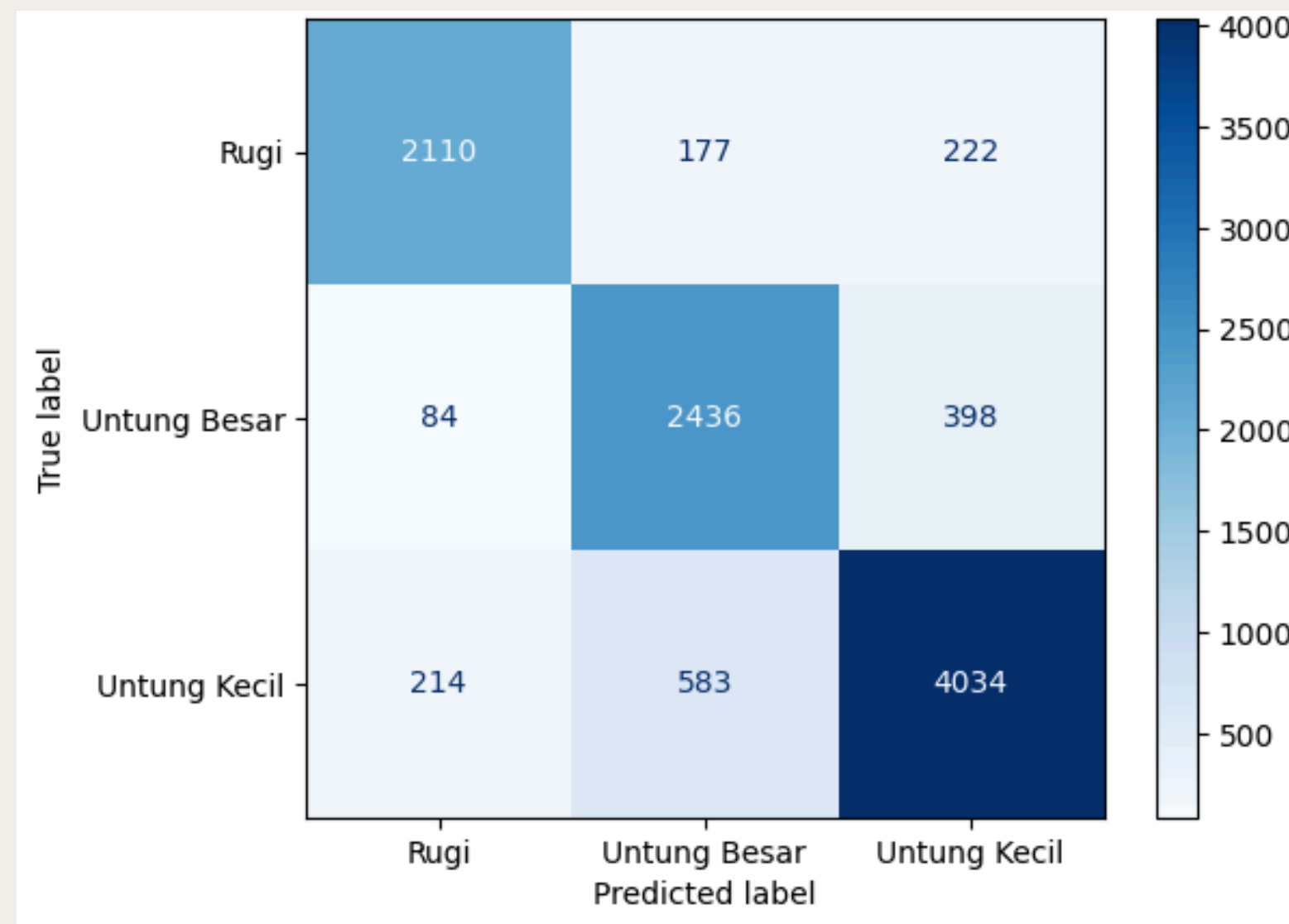
- DISCOUNT
- SALES
- CATEGORY
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- COUNTRY
- REGION
- QUANTITY
- SHIP_DATE
- CUSTOMER_NAME

Model Accuracy

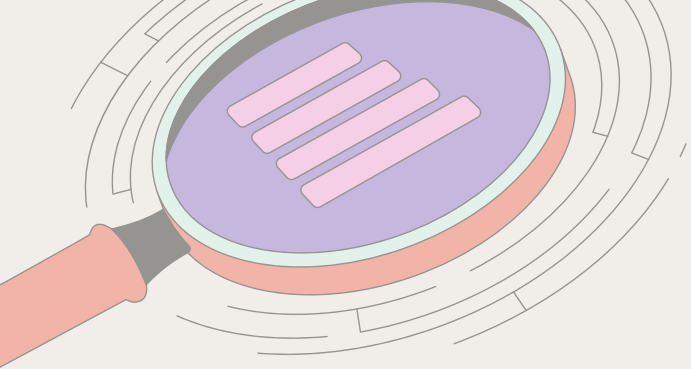
Model	Accuracy
Random Forest	84%
XGBoost + Optuna	84%
XGBoost	83%
StackingClassifier	83.4%
BaggingClassifier	82%
GradientBoostingClassifier	82%
VotingClassifier	77%
KNN	58%



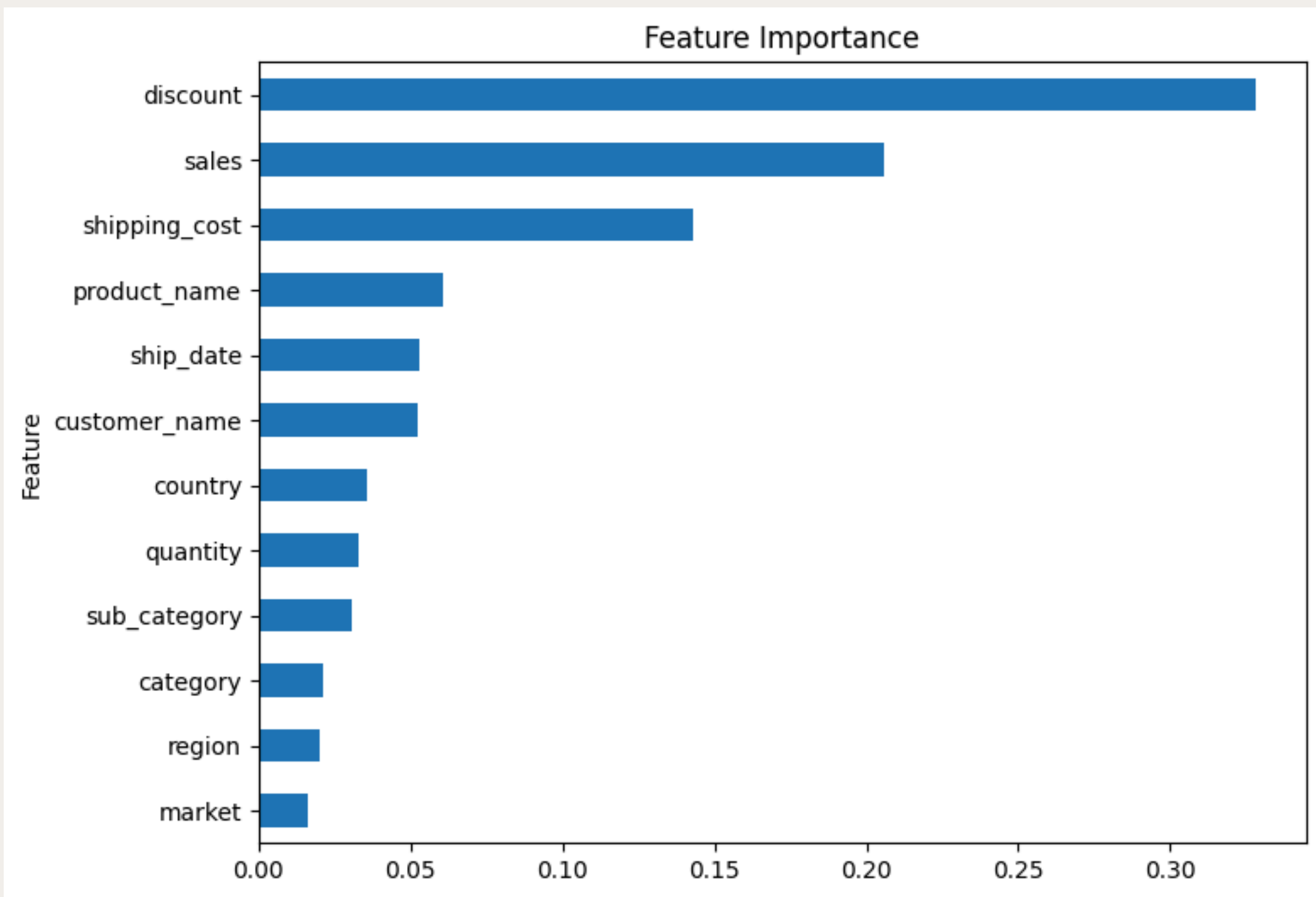
Random Forest



ACCURACY 84%
ROC AUC SCORE (OVR) 0.9562



Feature Importance by Random Forest

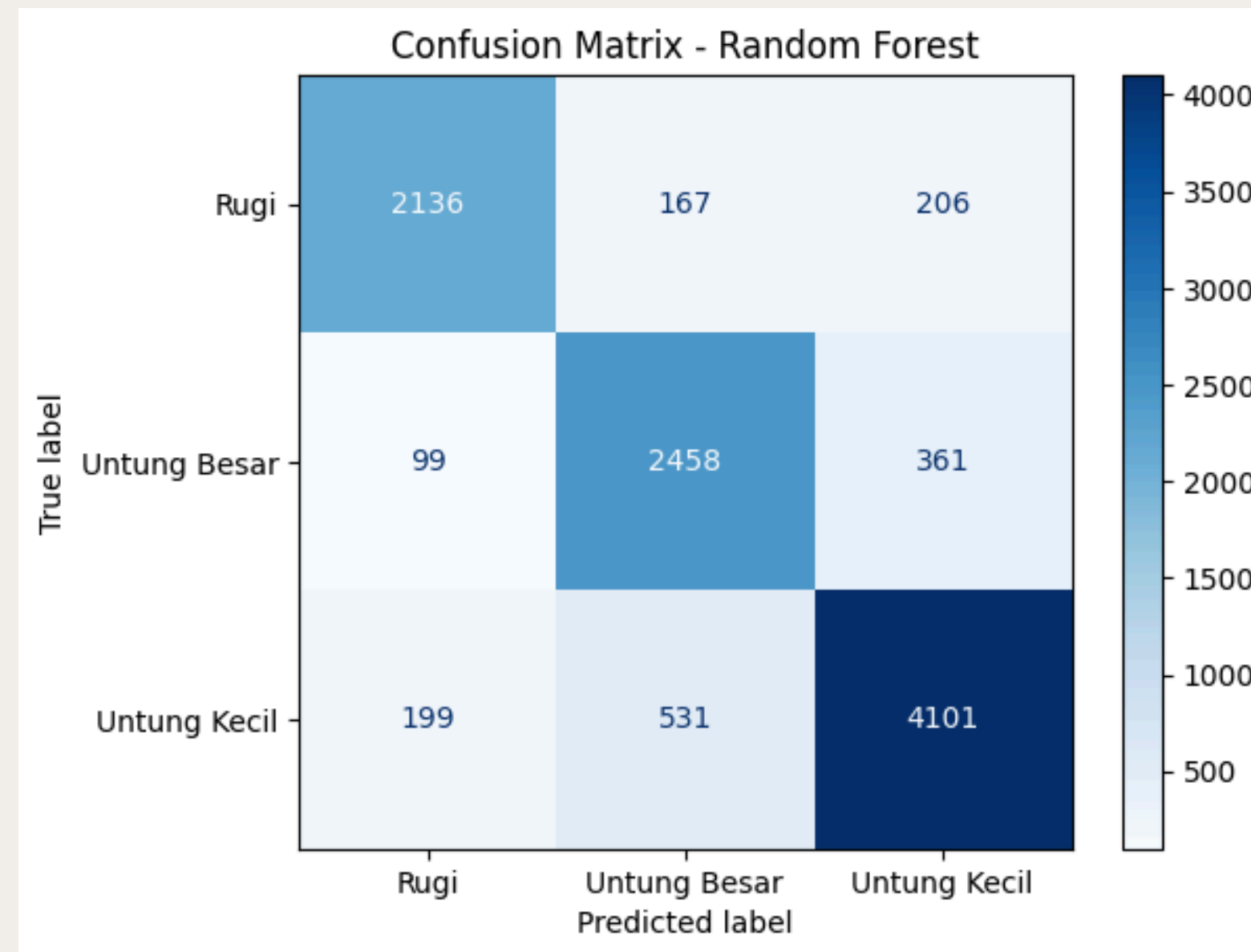


Feature	Chi2 Hitung
discount_bin	36065.6884
product_name	28422.4752
country	24149.4983
sub_category	10604.9665
category	6220.4128
ship_date	4812.2120
region	3323.9449
customer_name	2643.4157
market	1397.6546
order_priority	4.2271
ship_mode	3.1470
segment	1.0461

Model dengan 6 Feature

FEATURE

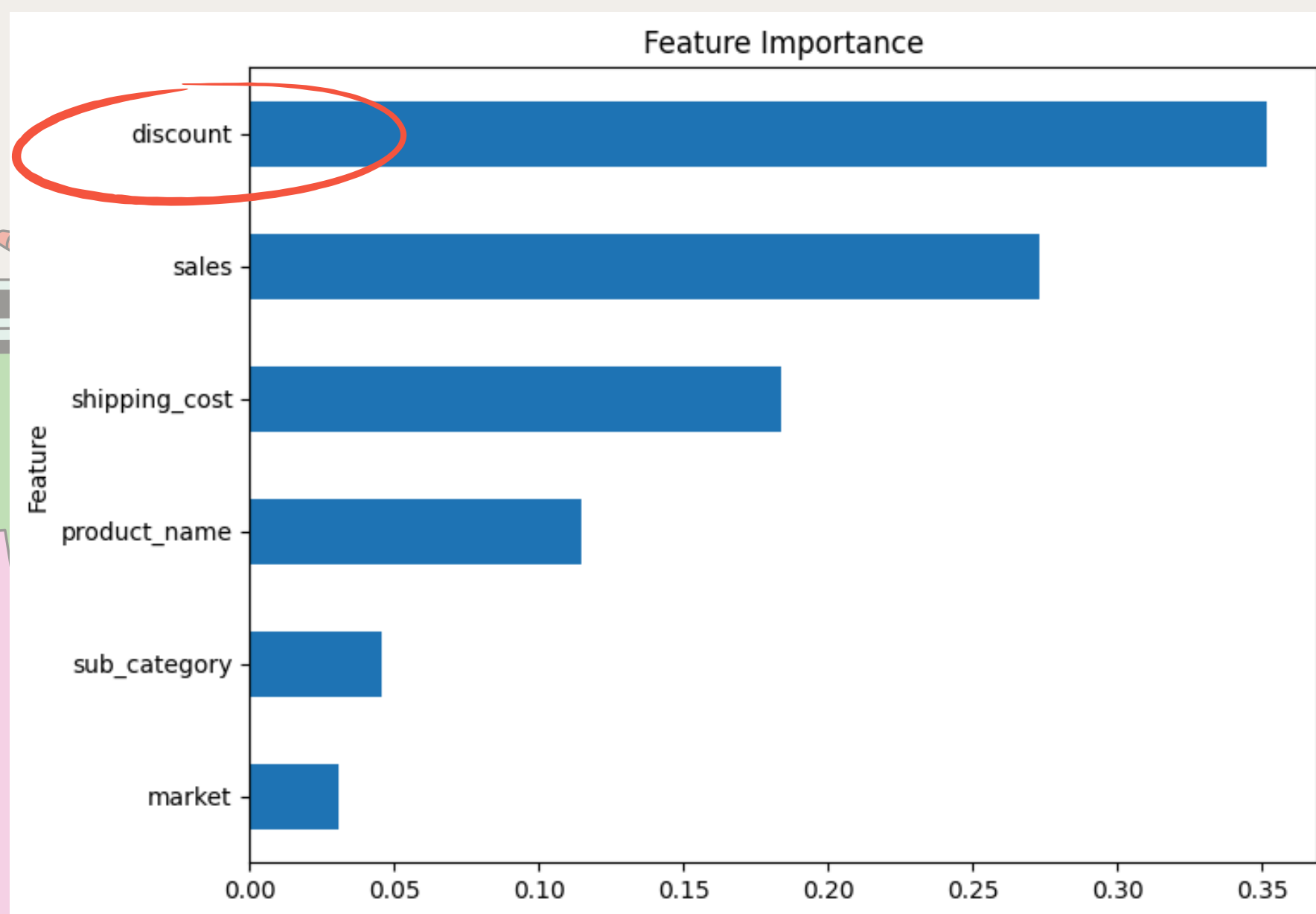
- DISCOUNT
- SALES
- MARKET
- SUB_CATEGORY
- SHIPPING_COST
- PRODUCT_NAME
- ~~COUNTRY~~
- ~~REGION~~
- ~~QUANTITY~~
- ~~SHIP_DATE~~
- ~~CUSTOMER_NAME~~
- ~~CATEGORY~~



ACCURACY 85%
ROC AUC SCORE (OVR) 0.9616

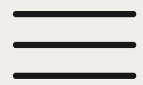


Model dengan 6 Feature



```
=== Feature Importance ===  
      Feature  RF Importance  
0      discount  0.351595  
1      sales    0.272951  
2  shipping_cost 0.184113  
3  product_name  0.115046  
4  sub_category  0.045771  
5      market   0.030524
```

Diskon masih menjadi variabel utama yang mempengaruhi profit

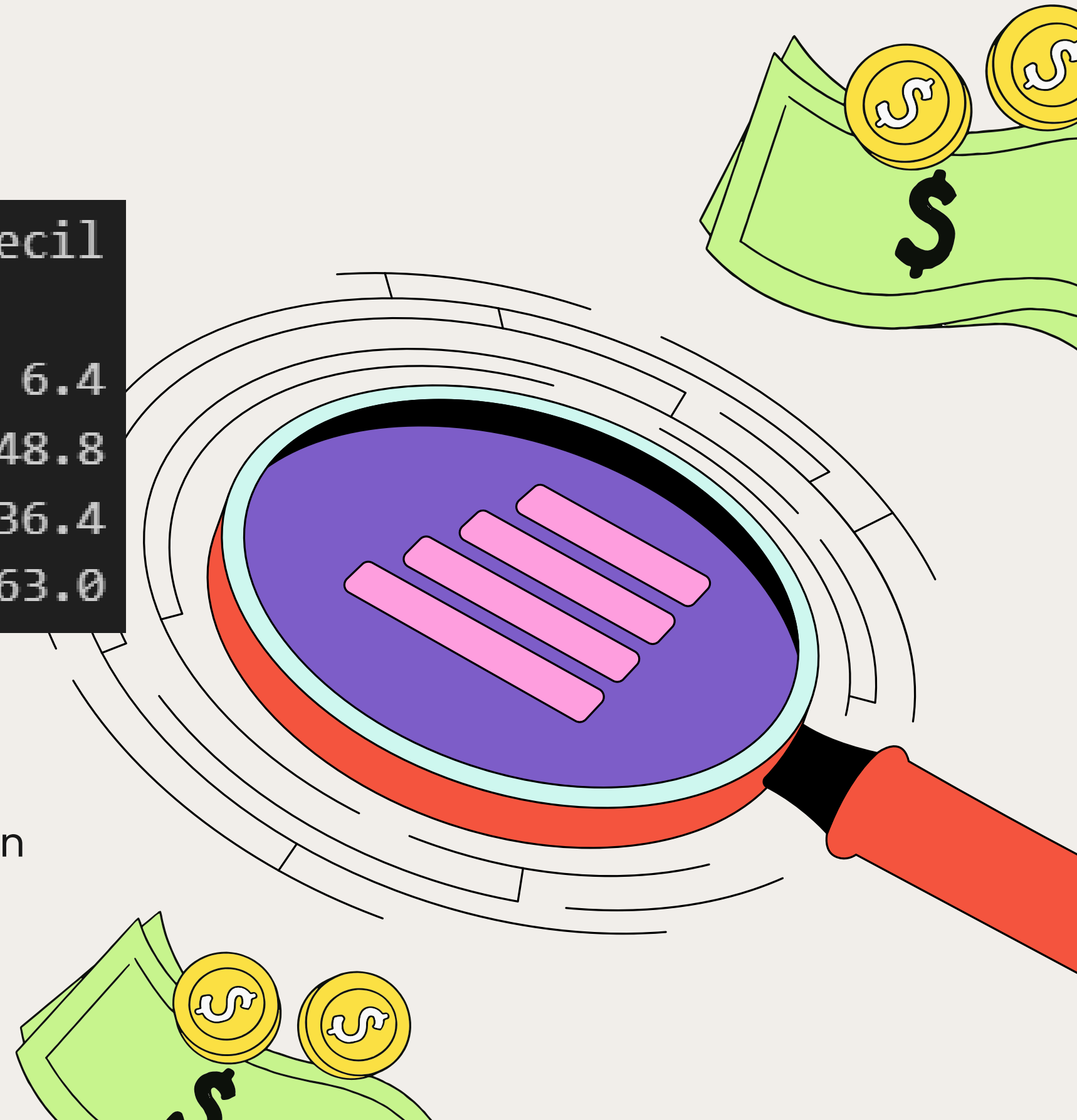


Analisis Lanjutan

profit_bin	Rugi	Untung Besar	Untung Kecil
discount_bin			
3	92.5	1.2	6.4
2	28.5	22.7	48.8
1	19.2	44.4	36.4
0	0.0	37.0	63.0

92.5% transaksi dengan diskon di atas 30% berakhir **Rugi!**
Hanya 1.2% yang Untung Besar, dan 6.4% Untung Kecil.

✓ Kesimpulan: Diskon terlalu tinggi justru menyebabkan kerugian besar. Ini sinyal bahaya untuk strategi pricing.





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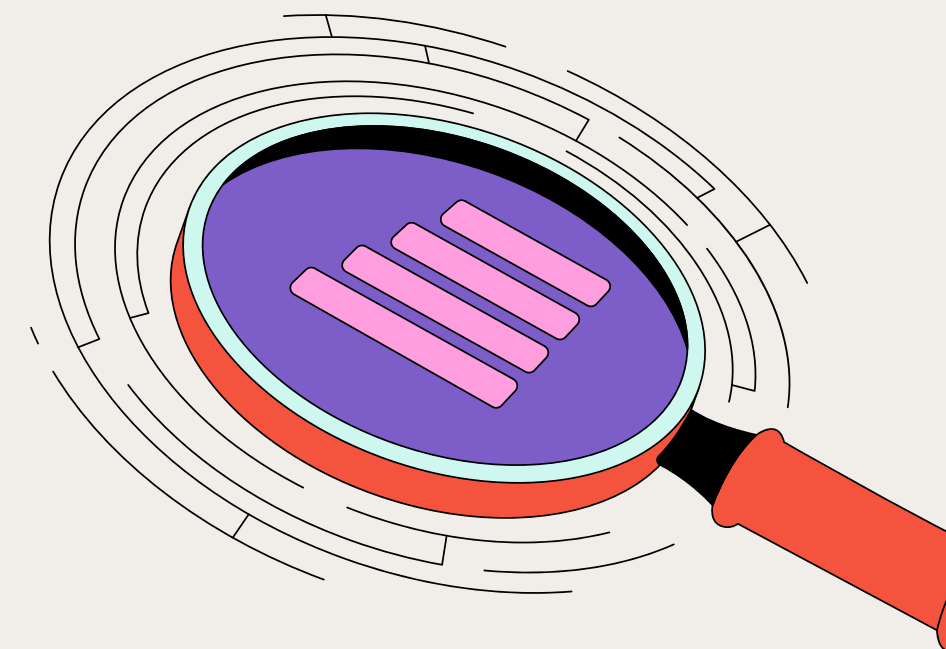


PERLU PENDEKATAN BARU

Optimalisasi Besar
Diskon

**TANPA MENGORBANKAN
MARGIN**

Maksimalkan diskon
kurang dari 30%





Kesimpulan

Accuracy model prediksi profit 85%

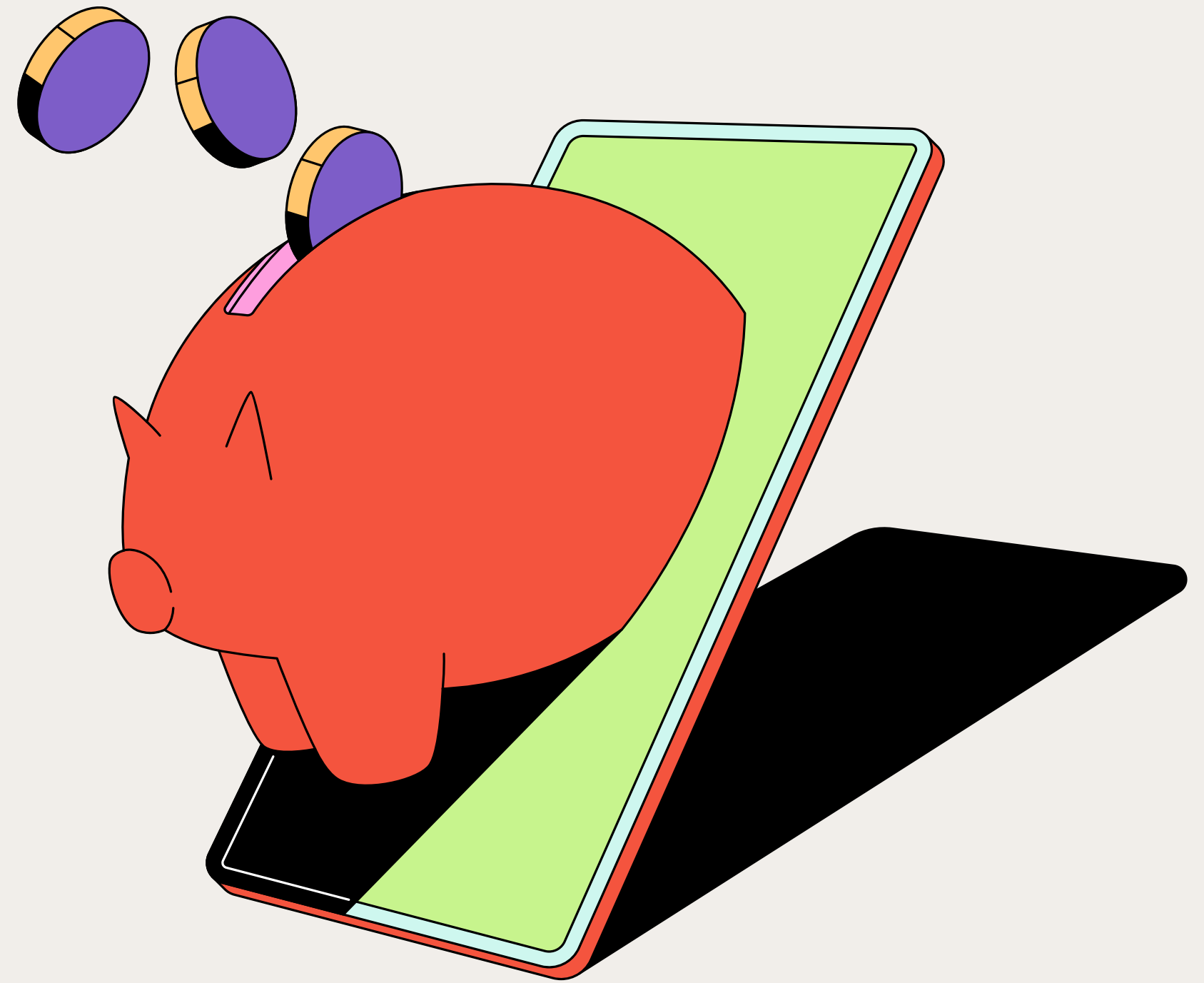
Diskon tinggi → justru meningkatkan beban rugi.

Evaluasi kembali pengaruh diskon terhadap sales

By Data, diskon kurang dari 30% menghasilkan profit yang cukup besar



Ide Tambahan



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Penerapan SLA (Service Level Agreement) untuk pengiriman



Customer akan mendapat
voucher discount hanya
apabila mengalami
keterlambatan pengiriman





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Benefit



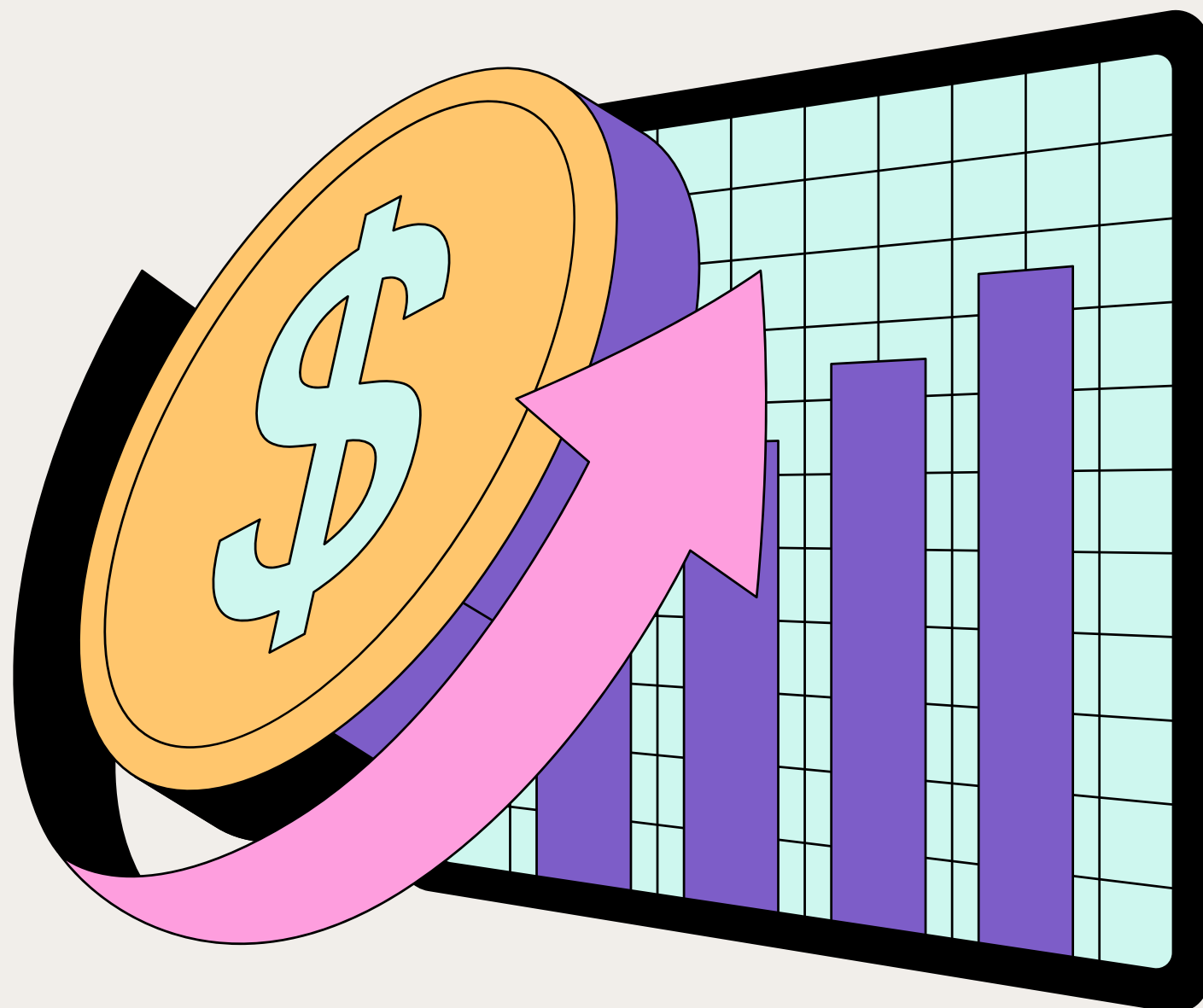
Cost Effective

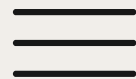


Menciptakan Repeat Order

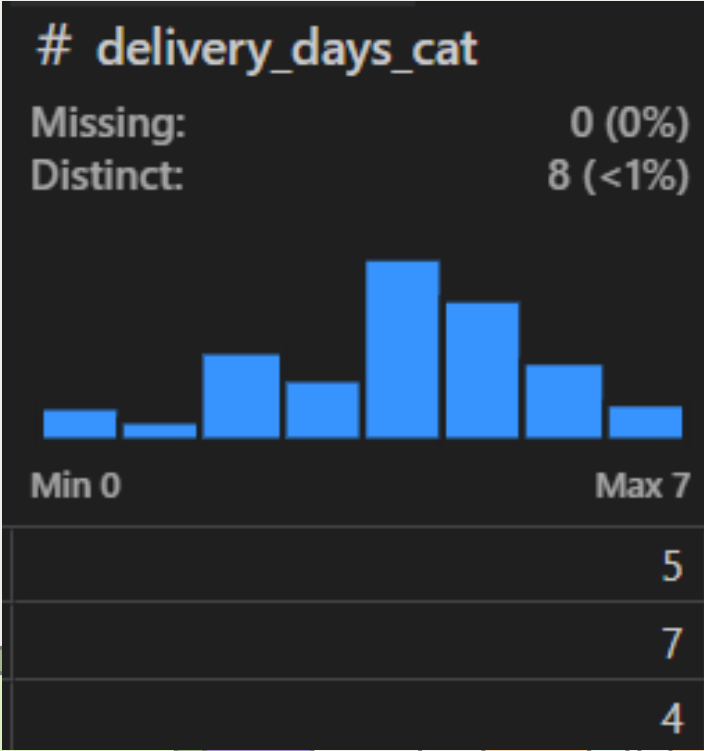


Meningkatkan Loyalitas Pelanggan





Membangun Model Delivery Speed



delivery_days_cat	
4 hari	14434
5 hari	11221
2 hari	7026
6 hari	6255
3 hari	5035
7 hari	3057
0 hari	2600
1 hari	1662



Chi Squared Test Of Independence

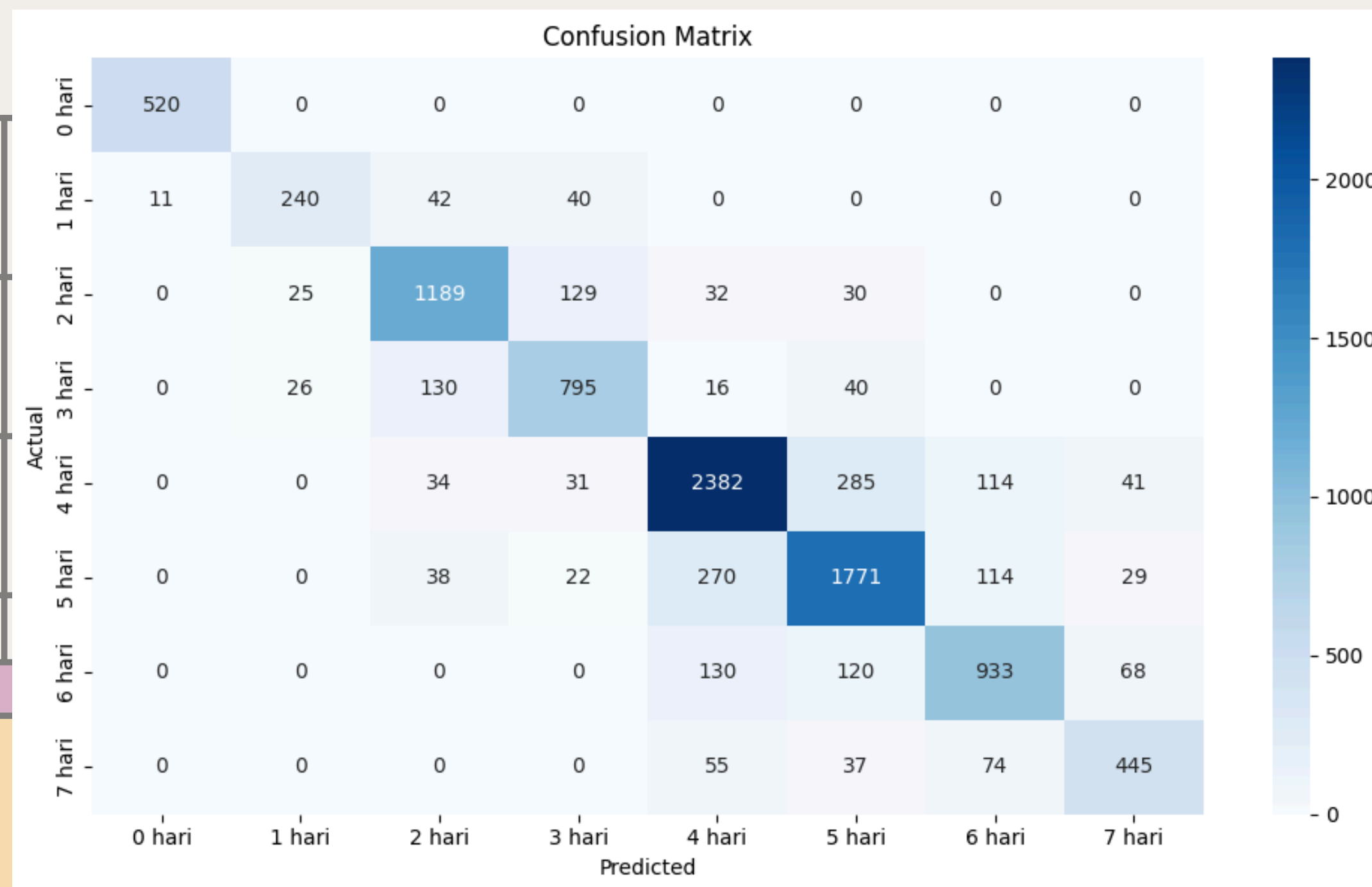
```
===== Hasil Uji Chi-Squared terhadap 'Delivery Speed' =====  
Feature      Chi2 Hitung  Chi2 Tabel  p-value      Kesimpulan  
segment      44.9272     23.6848     0.0000       ✓ Tolak H0  
region       319.3392    106.3948    0.0000       ✓ Tolak H0  
ship_mode    93450.7180   32.6706     0.0000       ✓ Tolak H0  
order_priority 30136.6393   32.6706     0.0000       ✓ Tolak H0  
country      2798.0755    1097.4844    0.0000       ✓ Tolak H0  
ship_date    38617.3511   10477.5353    0.0000       ✓ Tolak H0  
market       153.1353     58.1240     0.0000       ✓ Tolak H0  
customer_name 16127.6794   5732.5505    0.0000       ✓ Tolak H0  
order_id     350378.3895  176212.9057    0.0000       ✓ Tolak H0  
category      22.2720     23.6848     0.0732      Gagal Tolak H0  
product_name  26720.1248   26888.8717    0.1795      Gagal Tolak H0  
sub_category  122.4037     137.7015     0.2361      Gagal Tolak H0
```



Model Prediction

Accuracy: 81%

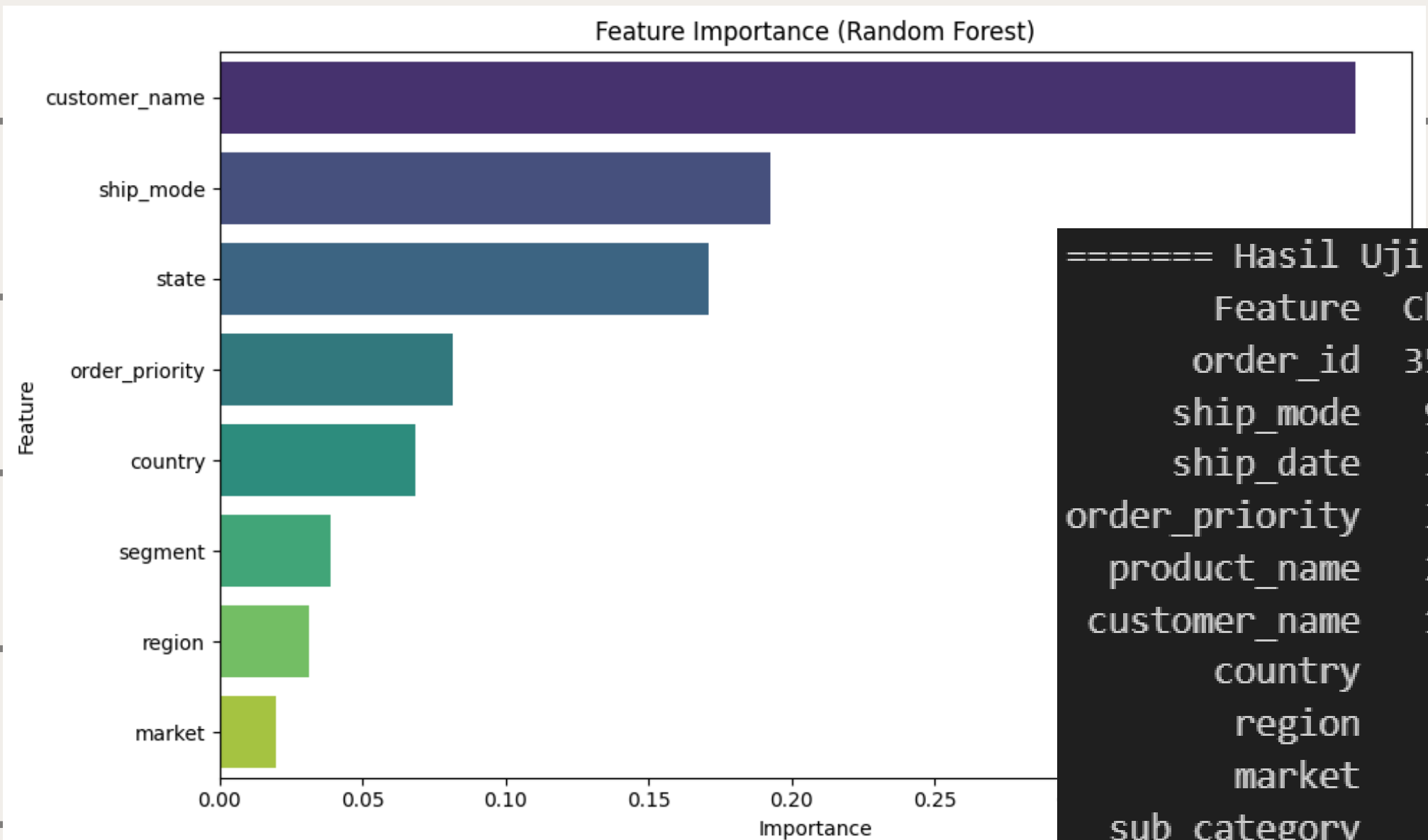
ROC AUC Score: 0.9665





Feature Importance

Accuracy: 81%



===== Hasil Uji Chi-Squared terhadap 'Delivery Speed' =====

Feature	Chi2 Hitung	Chi2 Tabel	p-value	Kesimpulan
order_id	350378.3895	176212.9057	0.0000	Tolak H0
ship_mode	93450.7180	32.6706	0.0000	Tolak H0
ship_date	38617.3511	10477.5353	0.0000	Tolak H0
order_priority	30136.6393	32.6706	0.0000	Tolak H0
product_name	26720.1248	26888.8717	0.1795	Gagal Tolak H0
customer_name	16127.6794	5732.5505	0.0000	Tolak H0
country	2798.0755	1097.4844	0.0000	Tolak H0
region	319.3392	106.3948	0.0000	Tolak H0
market	153.1353	58.1240	0.0000	Tolak H0
sub_category	122.4037	137.7015	0.2361	Gagal Tolak H0
segment	44.9272	23.6848	0.0000	Tolak H0
category	22.2720	23.6848	0.0732	Gagal Tolak H0



Kesimpulan

Accuracy model prediksi profit 85%

Accuracy model prediksi delivery speed 81%

Diskon tinggi $\neq$ loyalitas $\rightarrow$ justru meningkatkan beban rugi.

Penerapan voucher berbasis SLA memberikan win-win solution: biaya lebih terkendali, pelanggan tetap puas.





Action Plan

Evaluasi Kembali Diskon dan Model Machine Learning

PIC	Langkah	Deskripsi
Data/IT	Integrasi model prediksi profit & delivery	Gabungkan ke sistem operasional/sales pipeline
Marketing	Uji coba strategi SLA-voucher	Terapkan ke sebagian pelanggan high-priority
Finance & CS	Monitoring hasil	Evaluasi dampak ke margin dan retensi pelanggan
Management	Roll-out penuh	Ekspansi strategi ke semua area





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Work Together

Thank You For Your Attention

